

Wentao Chen

San Diego, California

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Github — Google Scholar

Research Interests

- **LLM Code Generation:** How can we systematically evaluate and improve the quality of LLM-generated code while effectively mitigating hallucinations?
- **Agent Memory Systems:** How can we design and manage efficient, scalable memory systems for LLM-based agents?
- **Time-Series Representation Learning:** How can we develop robust encoders that learn high-quality representations from time-series data with limited labeled samples?

Education

University of California, San Diego

Ph.D. in Computer Science

Co-advised by Prof. Rajesh K. Gupta and Prof. Jingbo Shang

San Diego, CA

2025 – 2030 (*Expected*)

Shanghai Jiao Tong University

B.E. in Computer Science

ACM Class, Zhiyuan Honor Program (Top 10%); GPA: 3.59/4.3

Shanghai, China

2021 – 2025

Publications

- Lizhe Zhang*, **Wentao Chen***, Li Zhong, Letian Peng, Zilong Wang, Jingbo Shang. "Memorize or Generalize? Evaluating LLM Code Generation with Evolved Questions."
- Binglin Zhou*, Linhao Zhong*, **Wentao Chen***. "Improve Cross-Architecture Generalization on Dataset Distillation." ICMSSP 2024.
- Keya Hu, Renjie Dai, **Wentao Chen**, Haolong Yin, Baoliang Lu *Fellow IEEE* and Weilong Zheng. "Contrastive Self-supervised EEG Representation Learning for Emotion Classification." EMBC 2024.

Experience

Shang Data Lab (SD Lab)

Visiting Student

San Diego, CA

2024.09 – 2024.12

- **Code Generation Memorization:** Investigated memorization phenomena in LLM code generation; proposed mitigation methods and evolved benchmarks.
- **LLM Attack Benchmark:** Evaluating diverse LLM attack methods across varied datasets using multi-dimensional metrics.

Center for Brain-like Computing and Machine Intelligence(BCMI)

Research Intern

Shanghai, China

2023.07 – 2024.02

- Researched self-supervised EEG representation learning for emotion classification.
- Optimized encoders to extract high-quality representations from limited labeled time-series data.

Selected Projects

Role-playing Drama LLM Optimization

python

2025

- Optimized drama LLM on agent memory system, proactive interaction and efficient finetuning method.
- Texas Hold'em Referee LLM**
python 2025
- Teached an LLM to perform as a referee for Texas Hold'em game only through prompt tuning.
- RISCV CPU Simulation**
Verilog, FPGA 2023
- Implemented a RISCv CPU featuring the Tomasulo algorithm, branch prediction, and out-of-order execution.
- Toy Compiler**
Java, Antlr4 2022
- Designed an end-to-end compiler for a C-like language, utilizing Antlr4 for CST generation and semantic analysis.

Teaching Experience

Shanghai Jiao Tong University	CS1951: Data Structure
<i>Teaching Assistant</i>	<i>2023.02 – 2023.06</i>
Shanghai Jiao Tong University	CS152: Programming (A)
<i>Teaching Assistant</i>	<i>2022.09 – 2023.01</i>

Skills & Awards

- **Programming:** C/C++, Python(skilled in pytorch, transformers, verl), Java
- **Communication:** TOEFL 99.
- **Awards:** Excellent Graduate of SJTU (2025); Zhiyuan College Honor Scholarship (2021-2024).